Topic Modeling in Financial App Negative Reviews

Research Question

How can NLP be used to identify and analyze UI/UX pain points in trading apps using negative App Store reviews?

Why Financial Apps

- **Rising Popularity:** Trading apps (Robinhood, Coinbase, Fidelity, Moomoo) have experienced significant growth in recent years
- **User Base Expansion:** Broader demographic adoption beyond traditional investors
- **Experience Challenges:** Despite popularity, users report numerous frustrations with usability, navigation, performance, and features

Why This Matters

For Users

- Poor UX can lead to costly mistakes and financial losses
- Frustration during market volatility amplifies negative experiences
- Learning curve creates barriers for new investors

For Companies

- User experience directly impacts retention and engagement
- App store reviews significantly influence download decisions
- Addressing pain points creates competitive advantage

Business Case for Sentiment/Topic Analysis

Product Improvement: Systematically identify product issues for development → Improve user satisfaction/retention

Competitive Advantage:

- Analyze competitor reviews to discover market gaps and opportunities
- Strategically differentiate app offerings based on user pain points

Reputation Management:

- Detect emerging bugs and usability issues
- Address negative feedback before it damages app reputation

Financial Apps



A comprehensive investment platform offering a wide range of financial services targeting traditional investors



An investment application recognized for its commission-free trading model and simplified user interface



A major cryptocurrency exchange platform, widely used for digital asset transactions and portfolio management



A mobile-based trading app with rich features, including advanced charts and analytical tools

Data Overview

- Collected using Python-based App Store scraper
- Random sample of 1,000 reviews per app
- Reviews span the last 5 years

coinbase_reviews_cleaned

date	developerResponse	review	rating	isEdited	userName	title	ml_rating
2021-10-11 01:28:16	{'id': 25674719, 'body': 'Hi there, thank you	impressed easy app use endless amounts information cryp	5	FALSE	dank0116	Easy money !	5
2025-01-23 03:06:29	{'id': 49919909, 'body': 'Hi there, Stevenra'	really good app crypto beginners intermediate users wish i	4	FALSE	Stevenra14	Great app, but missing a fe	4
2024-10-22 02:53:25	{'id': 47669014, 'body': "Hello Robinhooda	use coinbase three years happy service halfway year know	1	FALSE	Robinhoodallday	Used for years but has gor	1
2024-11-11 20:08:17	{'id': 48168788, 'body': "Hi CPG Tiberius, v	absolutely mind boggling unhelpful circular customer supp	1	FALSE	CPG Tiberius	Worst Customer Support	1
2025-04-12 14:35:28	{'id': 51660061, 'body': "Hi Usjnendin, we'r	left coinbase years ago kept removing payment methods la	1	TRUE	Usjnendin	Holding Money Hostage	1
2024-07-04 01:42:30		story goes like using coinbase everyday checking prices bu	1	FALSE	Vanzator	Can't get helped!!	1
2022-08-12 00:19:12	{'id': 31441366, 'body': 'Hi Caseyishappy, v	november th tried transfer btc another wallet told transactio	1	FALSE	caseyishappy	Lost $500	1
2022-01-31 21:22:16	{'id': 28397813, 'body': "Hi Mbev0391, we'	app simple investing recently decided get nfts wanted trans	1	FALSE	Mbev0391	Total disappointment	1
2022-01-04 01:40:46	{'id': 27746185, 'body': "Hi Cova13, thanks	first app used investing hopes customer service really brou	1	FALSE	Cova13	Disappointed	1
2023-02-09 00:42:19		coinbase loaded onto phone since early last week updated	1	FALSE	Zeus Monkey	App doesn't work with old	1
2022-01-20 22:36:04		welcome new consumer savings account consumer alone	5	FALSE	Doggs/doggerino/doggpounds	I am not a trader.	1
2024-03-23 21:21:15	{'id': 45298914, 'body': 'Hi there, we are sc	start saying since coinbase minimal issues giving stars ther	3	FALSE	Shmyizer	Missing functions	3

Methodology & Validation

Preprocessing

- Lowercasing
- Punctuation & emoji removal
- Stopword removal

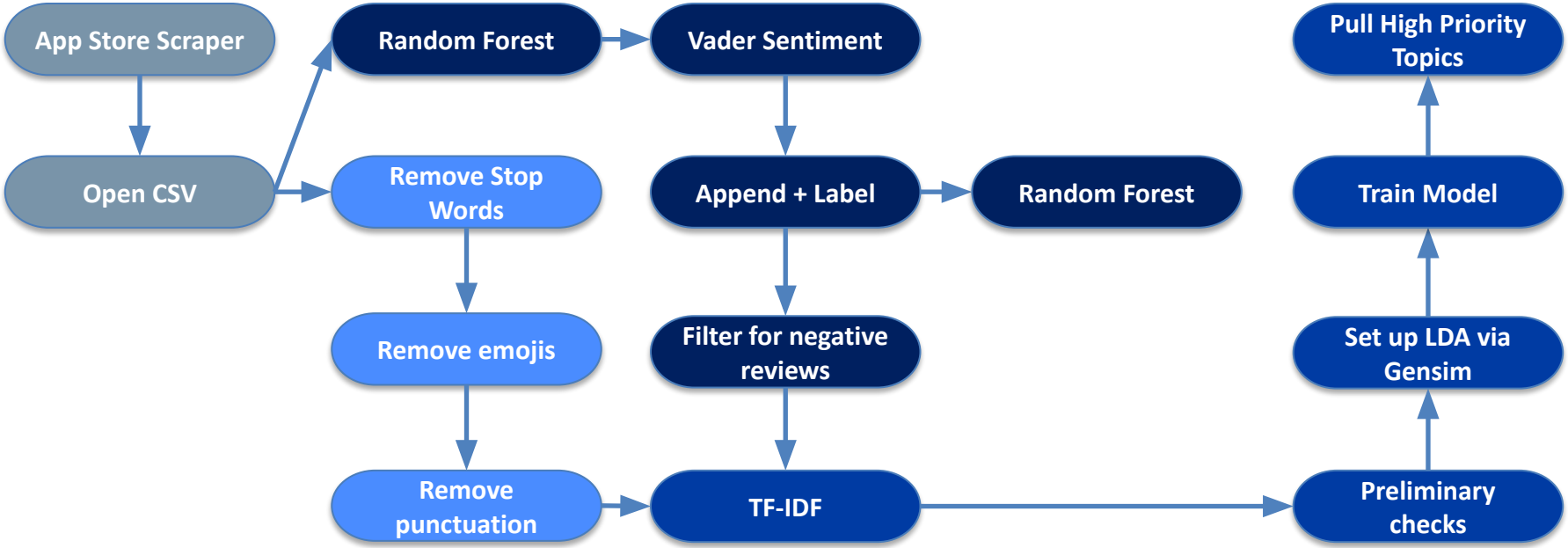
Sentiment Analysis

- Classify positive, neutral, negative reviews (VADER)
- Keyword extraction with TF-IDF
- Sentiment distribution across apps

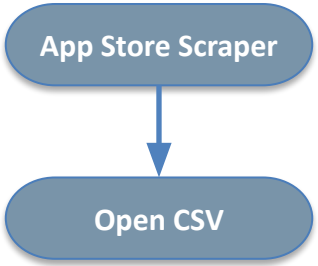
Topic Modeling

- Applied to **negative** reviews (LDA)
- Validation with spot checks with random reviews

Project Architecture



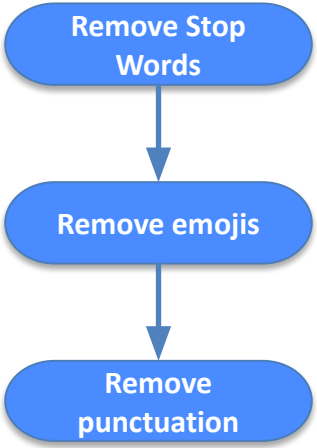
Project Architecture - Data Collection



Install & run App Store Scraper package

Open resulting CSV files of reviews

Project Architecture - Pre-Processing



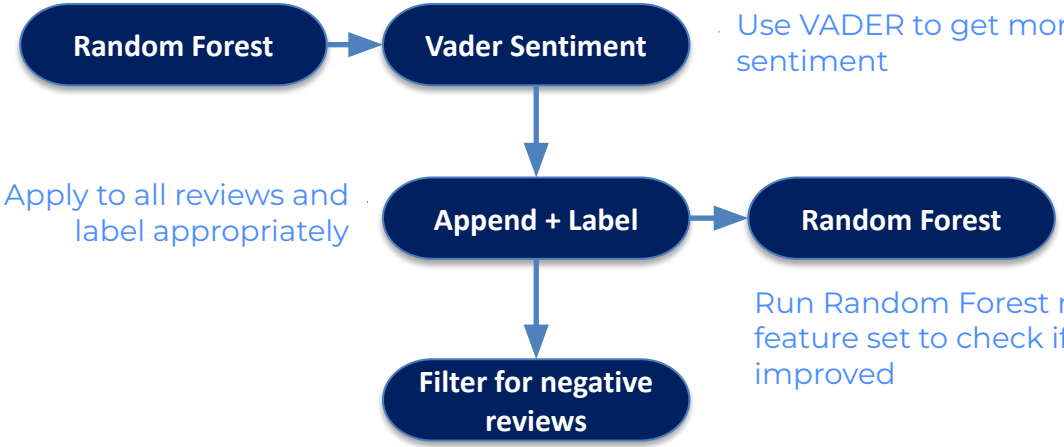
Remove stop words for topic modeling only

Remove emojis for topic modeling only

Remove punctuation except apostrophes for topic modeling only

Project Architecture - Sentiment Analysis

Run Random Forest model with TF-IDF to check if review text predicts rating



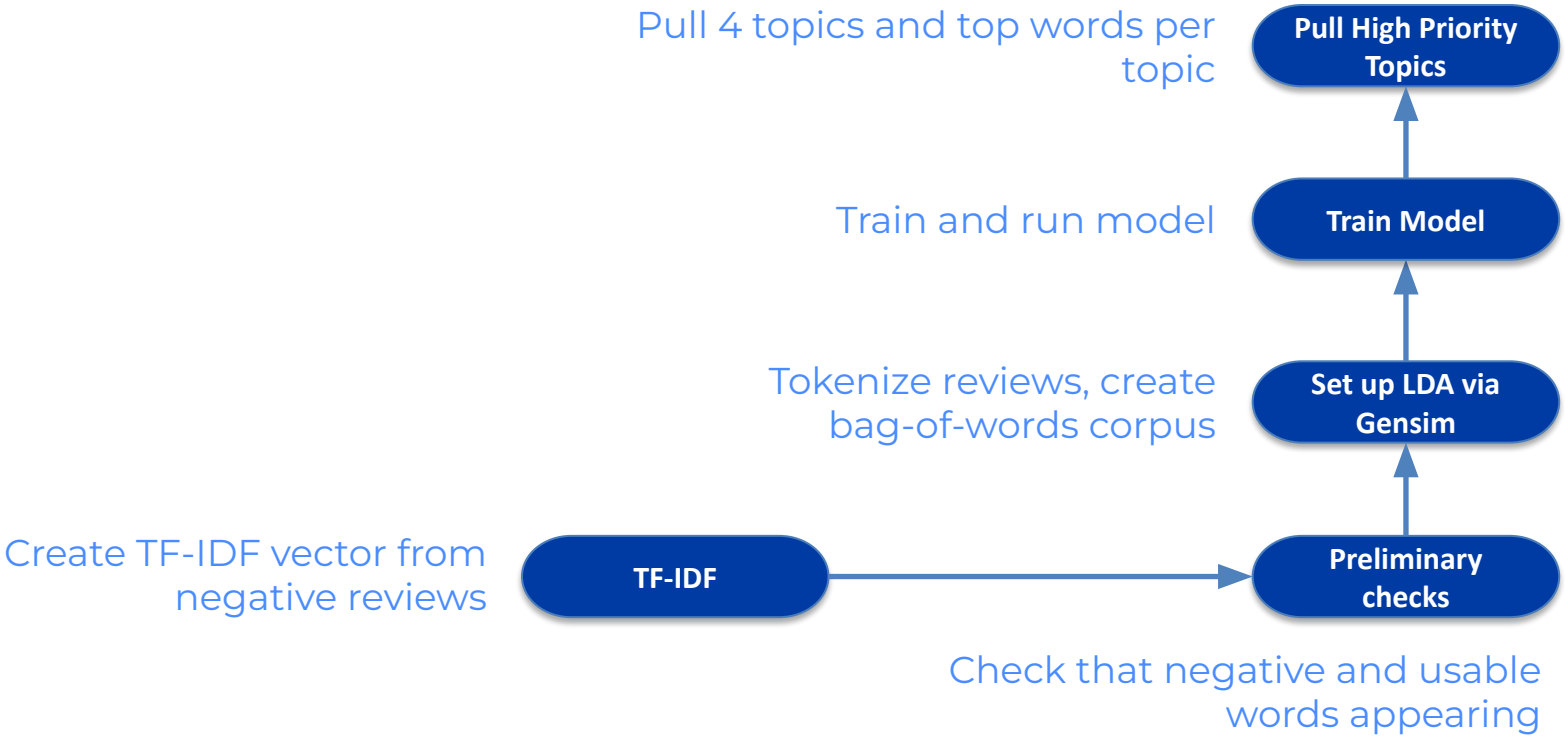
Use VADER to get more accurate sentiment

Apply to all reviews and label appropriately

Run Random Forest model with larger feature set to check if performance improved

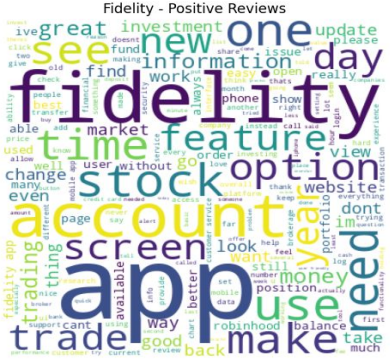
Filter dataset for reviews labeled “negative”

Project Architecture - Topic Modeling



Word Cloud Comparison

Fidelity



Robinhood



coinbase



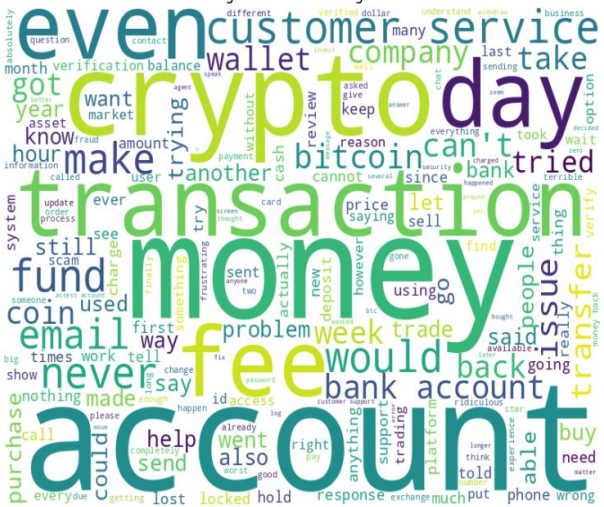
moomoo



From Unigram to Trigram

coinbase

Coinbase - Negative Reviews Unigram Word Cloud



Coinbase - Negative Reviews Bigram Word Cloud



Coinbase - Negative Reviews Trigram Word Cloud

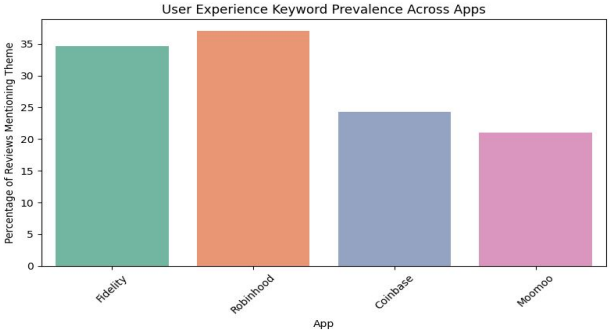


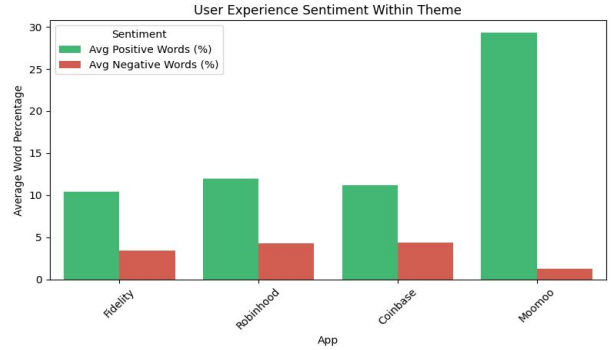
Main Topics in Negative Reviews

Fidelity	Robinhood	coinbase	moomoo
Topic 1: time, use, account Topic 2: money, account, funds, bank, days Topic 3: settle, dollar, verify, twice, week Topic 4: problems, spent, numbers, dollar, verify	Topic 1: personal, access, information, submit, automated Topic 2: account, money, app, stocks Topic 3: managers, institutions, family, id, exchange Topic 4: request, id, personal, automated, provide	Topic 1: account, money, bank Topic 2: hacker, department, deposits, withdrawal, lock Topic 3: repeatedly, statements, lock, drivers, deposits Topic 4: statements, lock, drivers, deposits, showed	Topic 1: service, customer, worst, got, app Topic 2: app, trading, order, open, sell Topic 3: money, deposit, platform, trade, account Topic 4: account, scam, phone, bank, withdraw

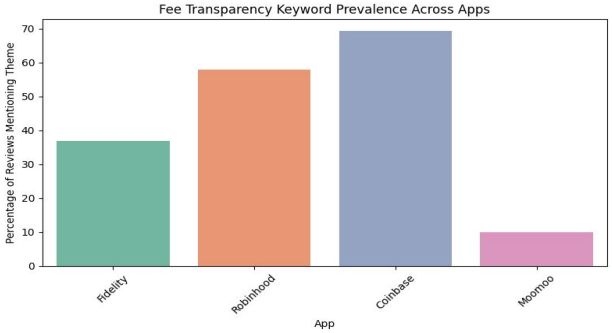
Topic Modeling

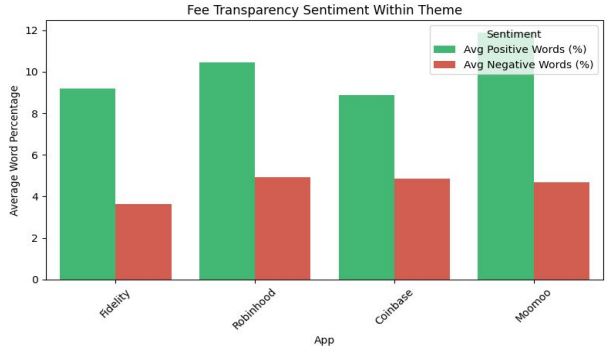
User Experience



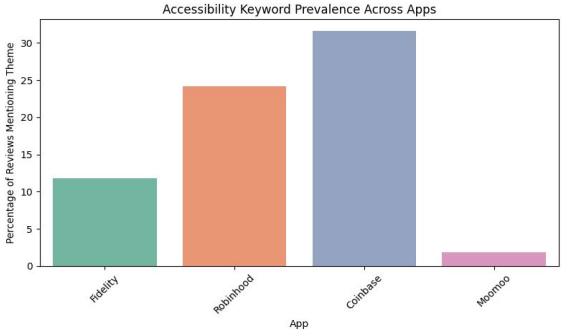


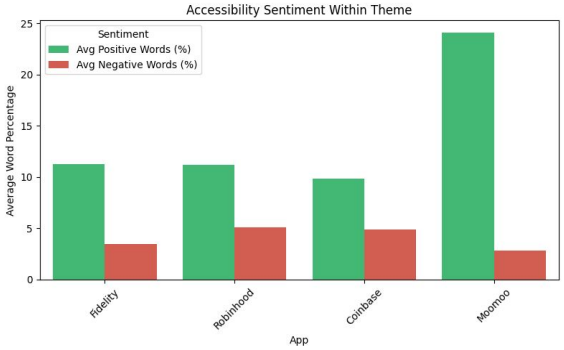
Fee Transparency



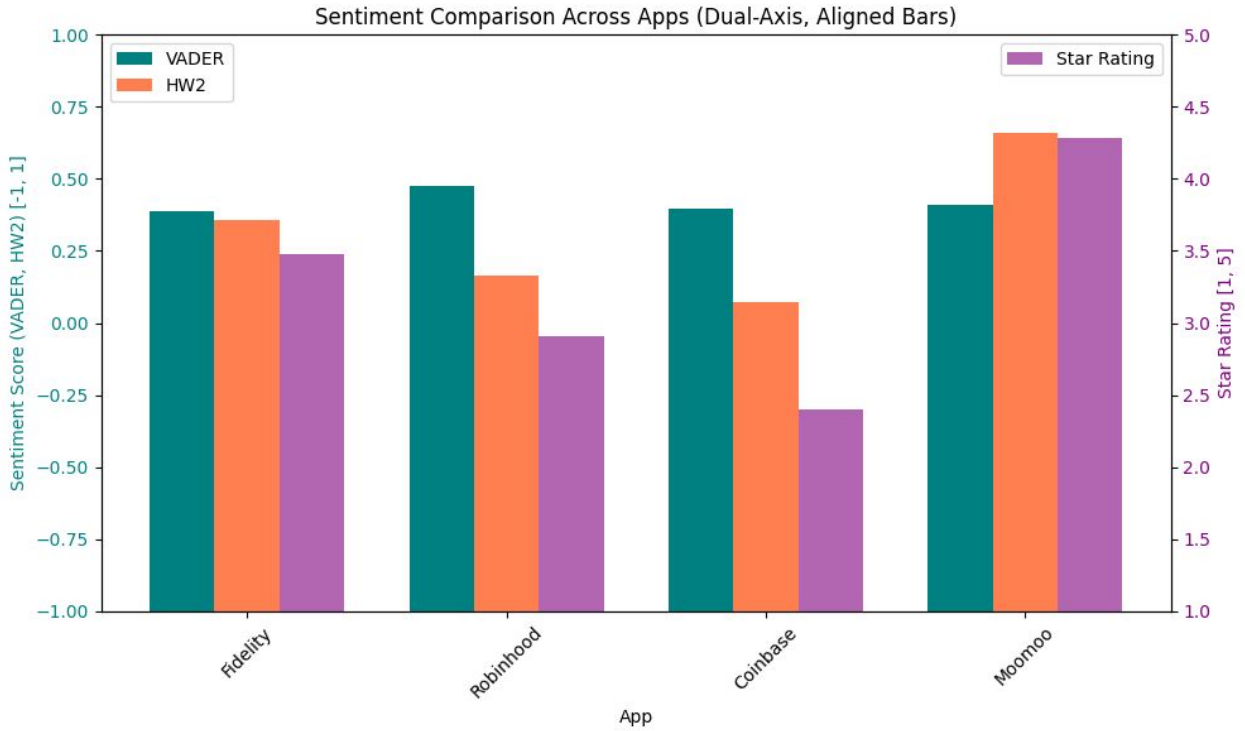


Accessibility





Sentiment Score Comparison



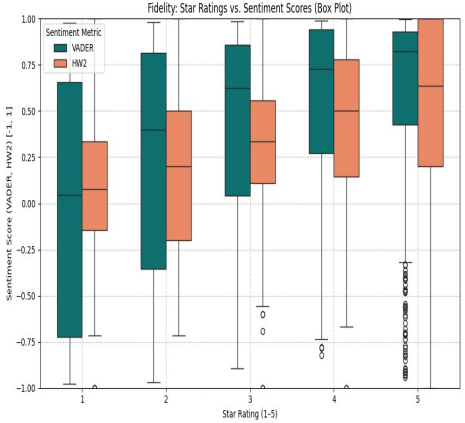
Star Ratings vs. HW2 and VADER Scores Box Plot

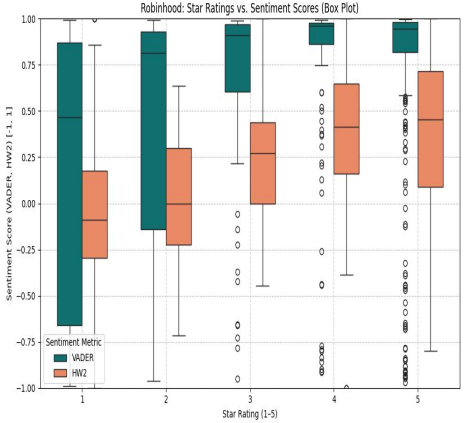
Fidelity

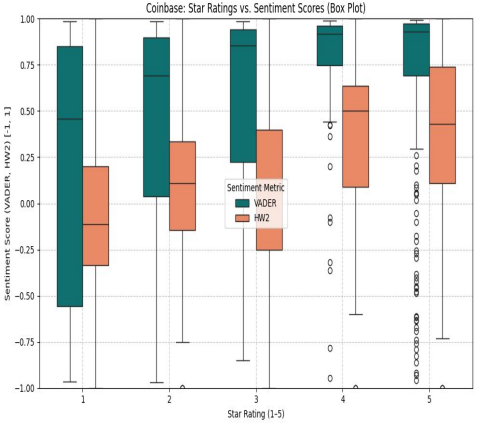
Robinhood

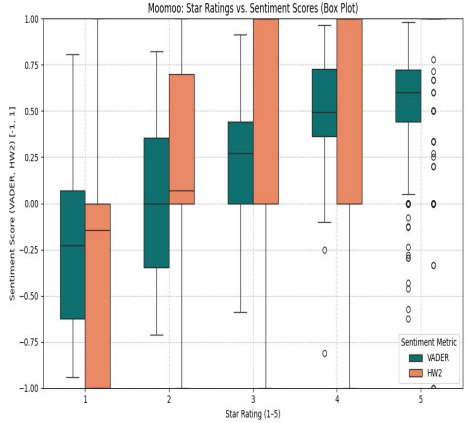
coinbase

moomoo









Random Forest Classifier

Data Splitting:

1. Split Fidelity data into 80% training and 20% testing sets
2. Use entire datasets for Robinhood, Coinbase, and Moomoo for evaluation

Features Used

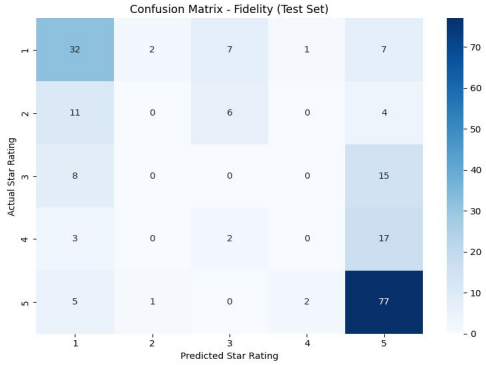
compound_score	VADER sentiment analyzer's compound score for the review (-1 to +1)
hw2_score	Custom sentiment score from gen_senti, based on normalized positive/negative word counts
review_length	Number of words in the review
pos_word_pct	Percentage of positive words (VADER score ≥ 0.05) in the review
neg_word_pct	Percentage of negative words (VADER score ≤ -0.05) in the review
sentiment_category	Numeric sentiment label (-1 for negative, 0 for neutral, 1 for positive)
avg_word_length	Average character length of words in the review
vader_hw2_diff	Difference between VADER compound score and HW2 score

Random Forest Classifier Result

Performance Metrics for Fidelity (Test Set):
Accuracy: 0.545

Classification Report:

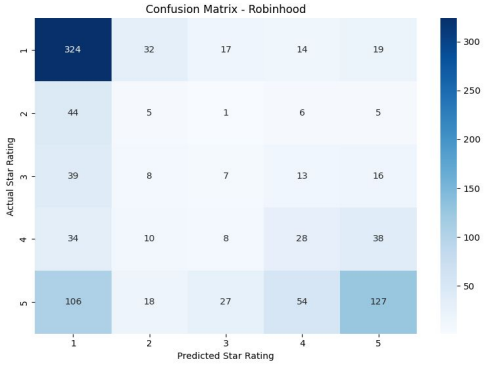
	precision	recall	f1-score	support
1	0.54	0.65	0.59	49
2	0.00	0.00	0.00	21
3	0.00	0.00	0.00	23
4	0.00	0.00	0.00	22
5	0.64	0.91	0.75	85
accuracy			0.55	200
macro avg	0.24	0.31	0.27	200
weighted avg	0.41	0.55	0.46	200



Performance Metrics for Robinhood:
Accuracy: 0.491

Classification Report:

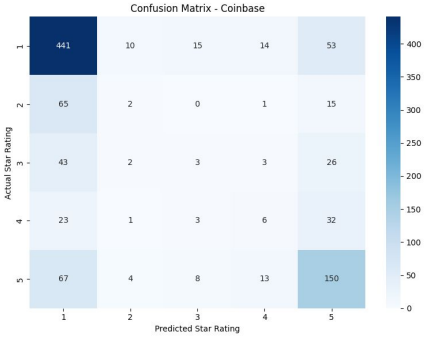
	precision	recall	f1-score	support
1	0.59	0.80	0.68	406
2	0.07	0.08	0.07	61
3	0.12	0.08	0.10	83
4	0.24	0.24	0.24	118
5	0.62	0.38	0.47	332
accuracy			0.49	1000
macro avg	0.33	0.32	0.31	1000
weighted avg	0.49	0.49	0.47	1000



Performance Metrics for Coinbase:
Accuracy: 0.602

Classification Report:

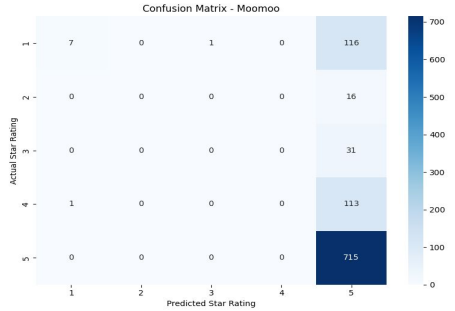
	precision	recall	f1-score	sup
1	0.69	0.83	0.75	
2	0.11	0.02	0.04	
3	0.10	0.04	0.06	
4	0.16	0.09	0.12	
5	0.54	0.62	0.58	
accuracy			0.60	
macro avg	0.32	0.32	0.31	
weighted avg	0.53	0.60	0.56	



Performance Metrics for Moomoo:
Accuracy: 0.722

Classification Report:

	precision	recall	f1-score	support
1	0.88	0.06	0.11	124
2	0.00	0.00	0.00	16
3	0.00	0.00	0.00	31
4	0.00	0.00	0.00	114
5	0.72	1.00	0.84	715
accuracy			0.72	1000
macro avg	0.32	0.21	0.19	1000
weighted avg	0.62	0.72	0.61	1000

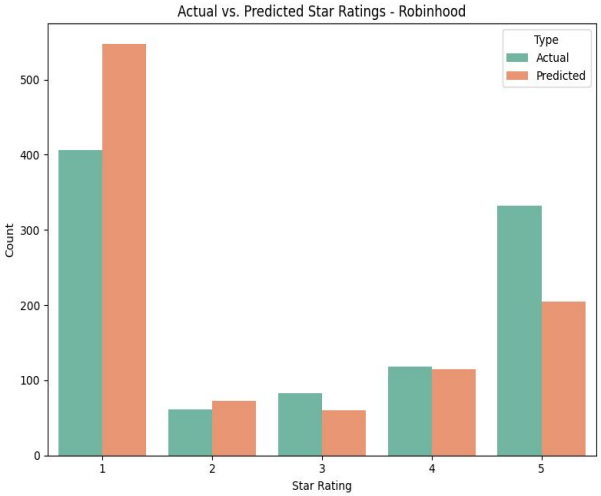


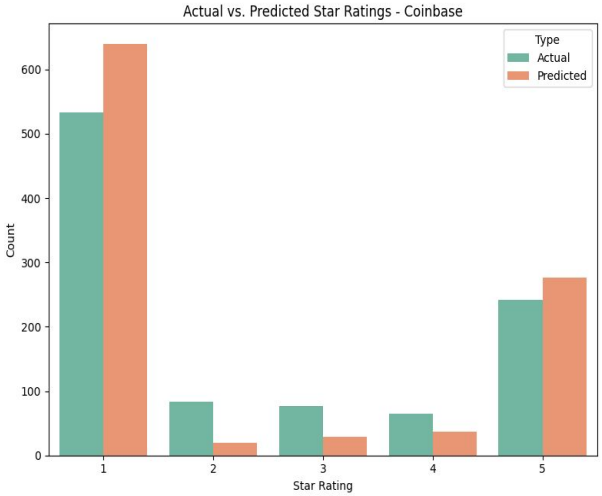
Actual vs. Predicted Star Rating

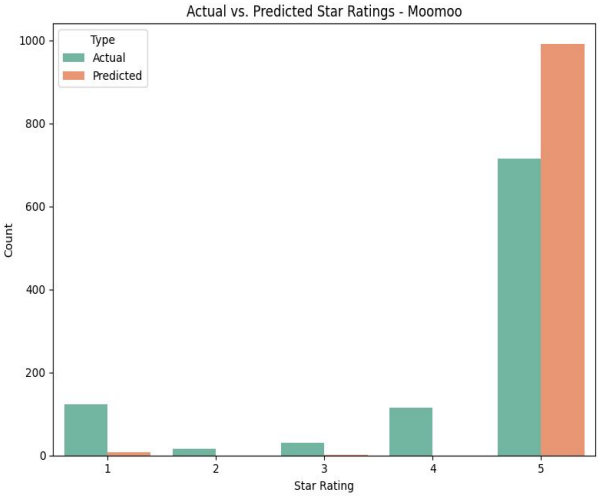
Robinhood

coinbase

moomoo







Pros and Cons Reflection

Pros	Data availability	Rich source data	Available literature	Replicability
Cons	Data complexity	Label inconsistency	Language variety	Misleading trends

Recommendations by App

Fidelity Investments

- Improve **user onboarding** and simplify **account verification**
- Streamline **buy/sell flows** to make trading easier
- Enhance communication around **fund movements**
- Focus on speeding up account setup, deposits, and trade execution

Coinbase

- Improve **customer support** (e.g., response times)
- Simplify **account recovery** and fund access (login issues, delayed withdrawals)
- Clarify and **communicate fees** upfront
- **Speed up** and stabilize transactions

Robinhood

- Improve **customer support** (unresolved issues, delayed help)
- Fix account and **withdrawal** issues
- **Market manipulation/distrust**: Rebuild trust in trading fairness
- Optimize **trading experience** (app lags, order execution delays)

Moomoo

- Improve **user onboarding and UI** for new traders, **customer support**
- **Streamline buy/sell flows** to reduce friction
- Improve **trust/transparency** around sign-up, deposits, account security
- Fix **withdrawals and transfers** delays

Conclusion



UI/UX cover many user pain points in these financial apps, though other issues such as trust in trading fairness and customer service response times are prevalent.



Star ratings don't always match the true sentiment of review text, requiring additional analysis to pinpoint truly negative reviews.



Natural language processing shows considerable promise in revealing high value features by making the voices of users both measurable and actionable.

Final Thoughts

Limitations

1. Subjectivity in interpreting topics
2. Bias in reviews (self-selection bias)
3. Moomoo had fewer reviews due to recent release

Future Direction

1. Track temporal changes in sentiment and pain points using time series analysis
2. Use BERT-based models for deeper sentiment extraction
3. Analyze sentiment tied to specific app features

References

Beginner's Guide to apple app store ratings & reviews. App Radar. (n.d.). https://appradar.com/academy/app-reviews-and-ratings/app-store-ratings-and-reviews

Egger, R., & Yu, J. (2022, May 6). *A topic modeling comparison between LDA, NMF, Top2Vec, and Bertopic to demystify twitter posts*. Frontiers in sociology. https://pmc.ncbi.nlm.nih.gov/articles/PMC9120935/

Isnan, M., Elwirehardja, G. N., & Pardamean, B. (2023, November 25). *Sentiment Analysis for TikTok review using Vader sentiment and SVM model*. Procedia Computer Science. https://www.sciencedirect.com/science/article/pii/S1877050923016800

Kumar Gupta et al. (2022, April 2). *Prediction of research trends using LDA based topic modeling*. Global Transitions Proceedings. https://www.sciencedirect.com/science/article/pii/S2666285X22000206



THANK YOU!